National University of Computer and Emerging Sciences 0016

Islamabad Campus

Numerical Computing (CS2008)

Course Instructor(s):
Dr. Muhammad Rizwan
Section(s): BS(CS) A 2022

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 50
Total Questions: 5

Date: Sep 23, 2025

22i-2372 — **Roll No**

A — **Course Section**

Student Signature

Do not write below this line.

Attempt all the questions.

CLO 3: [Demonstrate the capability of using numerical analysis library...]

Q1: **[2+8=10 marks]**

a.) Matrix A created via np.array([[32,64,86],[86,64,95],[55,55,66]]).

$$A = \begin{pmatrix} 32 & 64 & 86 \\ 86 & 64 & 95 \\ 55 & 55 & 66 \end{pmatrix}$$

What do the following two slices extract from the array? Write portion of A.

1. A[0:2, 2:3]
2. A[0:2, 1:2]

1. [86, 95]	2. [64, 64]

2

b.) Complete the table by writing the appropriate SciPy.Optimize functions names of the following mentioned numerical methods. It is required to fill in the table below and mention the core mathematical concept behind the method in simple mathematical or descriptive format.

Function	Parameters /Arguments	SciPy.Optimize function name	Output	Core Concept
Bisection	f, a, b, Tol=10^{-5}	bisec	x, f(x)	$f(a)\cdot f(b) < 0$, $\frac{a+b}{2}$
Fixed-point	f, g, x, Tol	Fixed_point	x, f(x)	$f(x)=0 \Rightarrow x^* = g(x)$
Newtons'	f, f', x, Tol	newton	x', f(x)	$x_{n+1} = x_n - \frac{f(x)}{f'x}$
Secant	f, xk, xk-1,	secant	x, f(x)	$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$

17/5

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[CLO 2: Analyze Sources of Error and behavior of different operators of numerical methods...]

Q2: (a) List and briefly explain the different types of errors that occur in numerical computation (with example) **[4.5+2.5+3=10 marks]**

Roundoff error	The method to expressing a number to its closest number e.g $\frac{1}{3}$ is displayed as 0.33 but its actual value is like ~~0.100~~ 0.33333....
Truncation Error	Method of assuming infinite number to a finite extent e.g $e^x = 1 + x^2 + \frac{x^2}{2!}$ it will give value to 3 decimal unit
Catastrophic Cancelation	Subtracting two similarly close value can result to ~~insi~~ very small remaining value $1.00002 - 1.00001 = 0.00001$

4.5

(b): If the relative error in measuring an area of 350m was 0.04, calculate the absolute error for that measurement.

$x = 350 \quad re = 0.04$

$$\frac{x - \hat{x}}{x} = 0.04$$

$$\Rightarrow \frac{350 - \hat{x}}{350} = 0.04$$

$\hat{x} = 336$

$ae = 350 - 336$

$\boxed{ae = 14}$

2.5

(c): Calculate the truncation error in calculating f'(2) for $f(x)=x^2$ by $f'(x) \approx \frac{f(x+h) - f(x)}{h}$ with h=0.2.

$$f'(2) = \frac{f(2.2) - f(2)}{0.2} = \frac{4.84 - 4}{0.2} \Rightarrow 4.2$$

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[CLO 1: Apply numerical methods for various mathematical operations...]

Q3: **[10 marks]**

Using the Bisection Method, perform three iterations to approximate the root of the following function within the interval [a,b]:

$$x^3 + x^2 - 1 = 0$$

with the initial interval [a,b]=[0, 2]
For each iteration, show the values of a, b, c=a+b/2, and f(c).
Finally, after three iterations, specify the new interval where the root of the function lies.

Iter#	a	b	c	f(a)	f(b)	f(c)
1	0	2	1	-1	11	1
2	0	1	0.5	-1	1	-0.625
3	0.5	1	0.75	-0.625	1	-0.015625
4	0.75	1	0.875	-0.015625	1	0.4355

after 3 iterations new interval is [0.75, 1]

10

[CLO 1: Apply numerical methods for various mathematical operations...]

Q4: Use Newton's method $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ to find a solution of the following, accurate to within 10^{-2}

$2x + 3\cos x - e^x = 0$, for $1 \le x \le 2$, by taking $x_0 = 1.5$ **[10 marks]**

$x_0 = 1.5$ $f'(x) = 2 - 3\sin x - e^x$

$x_{0+1} = 1.5 - \frac{1.151728}{-2.560219}$

$x_{0+1} = 1.5 + 0.449855$

$x_{0+1} = 1.949855 \approx 1.95$

$x_2 = 1.95 - \left(\frac{-0.1304}{-5.1307}\right)$

$x_2 = 1.95 - 0.2541$

$x_2 = 1.92$

$x_3 = 1.92 - \frac{0.0173}{-4.9214}$

$x_3 = 1.9235$

so, $x_3 - x_2 = 3.51 \times 10^{-3}$

which is more than required accuracy

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[CLO 1: Apply numerical methods for various mathematical operations]

7

Q5: Write a Python program to implement the Secant Method $x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$ **[7+3 marks]**

Define a function name my_secant that performs the root-finding procedure. Additionally, provide a separate code segment to test your implemented function

Code area for my_secant function to solve roots using Secant Method

```
def my-secant(f, xk, xk1, ToL, iter):
    i=0, err = xk1-xk
    while err < ToL or i <= iter
        err = xk1
        xk1 = xk1 - (f(xk1)*(xk1-xk)) ÷ (f(xk1) - f(xk))
        err = xk1-err
        i++
    return xk1, err
```

5

Write test code, which calls your method to solve the equation mentioned in Q4, with $x_0=1$, $x_1=1.6$, and $\varepsilon=10^{-2}$

```
f = lambda x: 2*x + 3*np.cos(x) - np.eps**2
x0 = 1
x1 = 1.6
tol = 10^-2
x, e = my-secant(f, x0, x1, tol)
```

2

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